Item: UVM-100 UV LED Optical Module | Process: Final assembly

PFMEA No.: PFMEA-001 | Rev: A | Date: 2026-11-05

Prepared by: O. Dhinoja | Core team: Mfg Eng, Quality, Design

Linked documents: Process flow OP10–OP60, Control Plan CP-001, ECO-001

Action threshold: any S ≥ 9 is reviewed; RPN ≥ 100 requires action.

Class: CC = critical (S ≥ 9), SC = significant (S 7–8)

Process Failure Mode and Effects Analysis: UVM-100 Module Assembly.

#	Op	Step	Failure mode	Effect	S	Class	Cause	Prevention control	O	Detection control	D	RPN	Recommended action	Owner / target	Revised S	Revised O	Revised D	Revised RPN
1	10	Inspect housing seat depth	Out-of-tolerance housing accepted	Lens at wrong height → focus shift, incorrect UV dose	9	CC	Machining variation, tool wear	Seat depth dimensioned directly 5.00 ±0.10 (Rev B, ECO-001)	3	100% check with GAG-100 flush-pin gauge	2	54	Reviewed: controlled by 100% GAG-100 check. No action	-	-	-	-	-
2	10	Inspect housing seat depth	Gauge gives false accept	Bad housing passes → incorrect UV dose	9	CC	Pin wear, debris in pin hole	Hardened A2 gauge (58–62 HRC), 0.055 min pin clearance	2	None (no gauge verification)	8	144	Verify gauge against a master setting part at the start of each shift; clean pin hole	Quality / 2026-11-15	9	2	3	54
3	20	Load housing in nest	Housing not fully seated	Board/lens installed misaligned	6	—	Debris in pocket	0.5 lead-in chamfer, Ø28.20 ±0.05 pocket	2	Visual: housing protrudes 5 mm above nest	5	60	None: below threshold	-	-	-	-	-
4	30	Install LED board	Board rotated 180°	LED off-axis → wrong UV beam	9	CC	Operator error	Key tab blocks wrong orientation (poka-yoke)	1	Board will not seat flat	2	18	Reviewed: designed out by key tab. No action.	-	-	-	-	-
5	30	Install LED board	Board damaged by ESD	LED degraded or fails in use	8	SC	Ungrounded operator or tools	ESD-safe acetal nest; ESD handling note on UVM-100	3	None (no electrical test)	9	216	Wrist strap + ESD mat; add a power-on light-output check at OP60	Mfg Eng / 2026-11-15	8	2	3	48
6	30	Install LED board	Board not flat on floor	LED tilted → beam off-axis	8	SC	Burr or debris on floor	Break edges 0.2 max; Ra 3.2 finish	3	Visual check	6	144	Clean floor (air blow-off) before placement; check board flat under magnifier	Mfg Eng / 2026-11-15	8	2	4	64
7	40	Place lens	Lens upside down	Wrong focus → incorrect UV dose	9	CC	Lens nearly symmetric	None	4	Visual check	7	252	Orientation mark on lens flat side (purchase spec); nest insert that accepts lens dome-up only	Design + Mfg Eng / 2026-11-30	9	2	3	54
8	40	Place lens	Lens tilted, not seated	Optical axis tilt → beam off-target	8	SC	Debris on ledge; lens catches counterbore edge	0.05 min clearance (ECO-001)	3	Visual check	6	144	Blow off ledge before placement; check lens rim flush and level	Mfg Eng / 2026-11-15	8	2	4	64
9	40	Place lens	Lens scratched or contaminated	Reduced UV output, scatter	7	SC	Bare-hand or tool handling	None	5	Visual check per WI-001 acceptance criteria (no scratches, fingerprints or particles under inspection light)	5	175	Finger cots and vacuum pick-up pen; inspect lens under light	Mfg Eng / 2026-11-15	7	2	3	42
10	50	Press retainer	Retainer not fully seated	Lens loose → shifts over time	8	SC	Uneven hand pressing	None (pressed by hand)	5	Visual check per WI-001 acceptance criteria (retainer flush and level on lens rim, no gap)	5	200	Design press tool (FIX-002) with hard stop for full seating	Mfg Eng / 2026-11-30	8	2	3	48
11	50	Press retainer	Lens cracked by press force	Loss of optical function; glass fragments	9	CC	Excess hand force	Interference limited to 0.01–0.05 by drawing	3	Visual check per WI-001 acceptance criteria (no cracks or chips under inspection light)	5	135	Same press tool with force limit; crack inspection under light	Mfg Eng / 2026-11-30	9	2	3	54
12	60	Final inspection	Defective module passed	Defect reaches user	9	CC	No defined acceptance criteria; inspector fatigue	Work instruction with photo acceptance criteria	3	Visual inspection only	6	162	Add power-on light-output check at OP60; boundary samples for visual checks.	Quality / 2026-11-15	9	2	3	54

Revised scores are projected; actions are planned, not yet implemented.

SEVERITY (S): how bad is the effect?

S	Criteria	Example (UVM-100)
10	Patient harm without warning; regulatory non-compliance	Uncontrolled UV exposure
9	Patient harm risk, or incorrect treatment	Incorrect UV dose (wrong focus)
8	Device does not perform intended function	Beam off-axis; lens loose
7	Degraded performance	Reduced UV output
5–6	Unit fails inspection, rework needed	Housing/board misaligned
3–4	Cosmetic defect	Scratched housing exterior
1–2	No noticeable effect	Minor internal burr

OCCURRENCE (O): how likely is the cause?

O	Criteria
1	Prevented by design (poka-yoke makes the error impossible)
2	Very rare; strong prevention control in place
3	Rare; prevention control in place
4–5	Occasional; depends on operator attention
6–7	Frequent; weak or no prevention control
8–9	Very frequent; easy mistake, no control
10	Almost certain

DETECTION (D): how likely does the failure escape?

D	Criteria
1	Error-proofed; cannot proceed if defective (part won't fit)
2	100% automatic or gauge check (go/no-go, flush-pin gauge)
3	100% measurement or functional test
4–5	100% visual inspection with defined criteria
6–7	Visual inspection without defined criteria, or sampling
8–9	Very unlikely to detect; no check in place
10	Cannot be detected

Rules:

S is not changed by controls; only O and D are.

The same effect gets the same S everywhere.

Prevention controls lower O; detection controls lower D.

Action threshold: any S ≥ 9 is reviewed; RPN ≥ 100 requires action.

RPN(Risk priority number) = S*O*D

Where S is Severity, O is Occurrence and D is Detection

Range: 1 to 1000. Higher the number, Higher the priority.

Note: RPN alone can hide high-severity risks (e.g. S = 9, O = 1, D = 2 gives RPN 18). Any S ≥ 9 is therefore reviewed regardless of RPN.